

UniNexus

An AI Academic Copilot for Smart University Management

Laravel 11 · Vue 3 + Inertia 2 · MySQL 8 (native vector store)

RAG + Agentic AI · Layered AI Proctoring · Pluggable AI Providers (OpenAI · OpenRouter · Jina · Mock)

One platform. Grounded AI. Every answer, provable.

1. Problem Statement

Modern universities still run on fragmented systems and manual workflows. Course materials, notices, assignment guidelines, attendance records and academic policies are scattered across ERP portals, messaging groups, classroom apps, spreadsheets and paper files, so students spend more time hunting for information than using it.

Faculty lose hours to repetitive work — attendance, authoring assessments, grading, writing feedback and chasing communication — and have no reliable integrity signal on online assessments. Administrators, meanwhile, have no live view of institutional health: problems such as disengaged or at-risk students surface only after the term has ended, when it is too late to intervene.

2. Proposed Solution

UniNexus is an intelligent, centralised academic platform powered by AI. Instead of forcing students, faculty and administrators across many disconnected tools, it unifies every major academic workflow behind one login with three tailored, role-based experiences — covering academic communication, course management, attendance, assessments, document knowledge, analytics and AI-assisted decision support.

The institution is modelled around a clear academic spine: departments contain courses, courses are offered as class sections each term, and students enrol into those sections. Everything else — materials, assignments, quizzes, attendance, exams and grades — belongs to the specific section a student is actually in, so each user only ever sees content relevant to their own classes.

The core innovation is a **Retrieval-Augmented Generation (RAG)** architecture. Unlike generic chatbots that hallucinate, UniNexus grounds every answer in institution-approved documents — syllabi, policies, handbooks and lecture materials — and also in the user's own notes and their sections' course materials ("chat with my materials"). Every response streams token by token and carries source citations (down to the document and page) plus a confidence score, making the system trustworthy and auditable.

Beyond answering, the assistant **acts**. The student tutor is agentic: through eight permission-checked tools it can check upcoming deadlines, list courses, browse and book faculty office hours, generate practice quizzes and flashcard decks, and add planner tasks — showing a live tool-activity trail in the conversation. A segmented Agent / Answers-only switch lets the student disable tools entirely, enforced server-side.

Multi-provider AI with automatic fallback: UniNexus runs on pluggable AI providers — OpenAI (gpt-4o + embeddings), OpenRouter (an OpenAI-compatible chain of free models), Jina (free multilingual embeddings) and a built-in deterministic Mock provider that needs no API key at all. Chat and embeddings each resolve through fallback chains that end in the never-failing Mock provider, vectors are model-tagged so providers never cross-compare, and switching providers automatically re-embeds the knowledge base — so the entire product, including streaming AI chat, generation and agentic tools, can be demonstrated with zero external setup or cost.

Assessment integrity is protected by **layered, AI-assisted proctoring**: seventeen independent security layers — from fullscreen enforcement and randomisation up to webcam/screen recording and three on-device camera-AI layers (a blink-verified face-liveness gate, phone and second-face detection, and snapshot photo evidence) — selectable per test by faculty from an admin-governed set, with every attempt producing a risk-scored review dossier.

3. Objectives

- Provide students, faculty and administrators with a single unified platform for all academic activity, eliminating the need to switch between disconnected tools.
- Deliver accurate, hallucination-free AI responses by grounding every answer in verified institutional documents and the user's own materials, with citations and confidence scores.
- Reduce repetitive faculty and administrative workload by automating quiz and assignment generation, rubric grading drafts, attendance, feedback summarisation and announcements.
- Move the assistant from answering to acting — letting students complete real academic tasks (booking office hours, generating practice, planning study) directly in conversation, under the same permission checks as the UI.
- Safeguard assessment integrity with layered, privacy-conscious proctoring — on-device camera AI, behaviour logging and risk scoring that produce review signals for humans, never automated verdicts.
- Enable real-time institutional visibility through centralised analytics, user-activity reporting, AI-usage monitoring, early at-risk detection and governed document management.
- Build a permission-aware, role-based AI system where every user — student, faculty or admin — only accesses information authorised for their role.
- Guarantee the system runs anywhere by shipping resilient multi-provider AI with automatic fallback — including a fully offline, keyless provider — removing any hard dependency on external services, and give operators safe public-demo controls (AI usage caps and an instant maintenance switch).

4. Methodology / Technology Used

UniNexus is a Laravel + Inertia monolith with a domain-driven service layer and an AI pipeline designed for reliability, auditability and low operating cost.

- **RAG pipeline**: approved documents are chunked into overlapping passages, embedded as vectors, and retrieved by cosine similarity (top-K) against the embedded user query; retrieved passages are supplied to the model as grounded context. Retrieval scope is the union of library documents the user may see, their own notes, and the materials of their enrolled or taught sections — so two people asking the same question may receive answers built from different sources.
- **Agentic layer**: the provider contract accepts function-calling schemas; a bounded agent loop (max 3 rounds, student chat only) executes requested tools through the real domain services, so RBAC, section scoping and atomic booking bind AI actions exactly as they bind UI clicks.
- **Backend**: Laravel 11 (PHP 8.2+) with layered authorisation — role and permission middleware, permission-aware form requests, and per-record policies — over a DDD-flavoured domain layer (User, Academic, Chat, RAG, Search, Notification, Analytics, Community).
- **Frontend**: Vue 3 with Inertia 2, Vite and Tailwind CSS — a single-page experience with no separate API client, role-differentiated per login; SSE delivers token-by-token streaming.
- **Database & vector search**: MySQL 8 holds structured academic data (users, courses, sections, attendance, assessments, document metadata) and the document embeddings, with cosine similarity computed in-application — no separate vector database to deploy, operate or pay for.

- **AI providers & models:** OpenAI gpt-4o for reasoning, tool calling and vision (OCR of handwritten notes) with text-embedding models for retrieval; OpenRouter as an OpenAI-compatible free-model chain; Jina for free multilingual embeddings; and a deterministic, keyless Mock provider (streaming- and tool-capable). Chat and embeddings resolve through separate fallback chains ending in the Mock provider; vectors are model-tagged and re-embedding is dispatched automatically on provider switch. All AI parameters — provider, models, creativity, passages per question, minimum relevance threshold, tone and supported answer languages — are admin-configurable.
- **Exam security:** a config-driven registry of 17 independent proctoring layers resolved per test as config default → admin global gate → faculty selection; the camera-AI layers (face liveness, phone detection, snapshot evidence) run entirely on-device in the browser, with recordings and snapshots stored privately and pruned on a retention schedule.
- **Operations:** a zero-latency visit-tracking middleware feeds an admin user-activity report (visitors, referrers, devices, IP-derived locations); a demo-mode cap meters AI requests per account across every AI surface; and a hidden URL-driven maintenance switch can freeze or restore the whole app instantly.
- **Chat modes:** academic support, exam preparation, research, assignment help, career guidance and general — each injecting a different system prompt — with a selectable answer language (7 supported).

5. Expected Outcomes

- A significant reduction in the time students spend searching for academic information — materials, policies, notices, their own notes and course content are all reachable through one cited, streaming AI interface, plus a semantic  search across everything they can access.
- Decreased faculty workload on repetitive tasks: AI-generated quizzes and assignments, AI-drafted per-criterion rubric grades, AI-summarised course feedback and one-click attendance return hours to teaching.
- Higher accuracy and trustworthiness of academic information, as answers are grounded in approved documents rather than generic internet data, and carry citations and confidence.
- Credible online assessment: layered proctoring with on-device camera AI, evidence trails and risk-scored dossiers makes remote class tests trustworthy while keeping humans in charge of every verdict.
- Fewer missed deadlines and better study habits, through agentic in-chat actions, the AI Study Planner, spaced-repetition flashcards, a concept mastery map and weekly email digests with deadline nudges.
- Earlier intervention for struggling students, as faculty see live at-risk flags on four signals (attendance, missed deadlines, test average, grade) instead of discovering problems at term's end.
- A measurable reduction in academic miscommunication, as notices, policies and announcements flow through one verified, centralised platform.

6. Core Features

Students

Streaming, cited AI tutor grounded in the library and in their own notes and course materials; agentic in-chat actions (8 tools) with an Agent / Answers-only switch; self-serve AI practice quizzes and question-bank self-quizzing; flashcards with SM-2 spaced repetition; OCR of handwritten notes into searchable, AI-indexed notes; AI Study Planner; "My Progress" analytics with a concept mastery map and one-click adaptive review; prerequisite badges and waitlist positions at registration; anonymous peer review and anonymous course feedback; group study rooms; office-hours booking; discussion feeds; an opt-in gamified leaderboard; attendance, GPA/CGPA, transcript, degree roadmap, unified calendar with .ics export and subscribe; notes, tasks and real-time messaging with their faculty.

Faculty

Streaming AI teaching assistant for quizzes, assignments and rubrics; a per-course shared question bank (manual add or import from past tests, spun into draft class tests); timed class tests with layered, per-test proctoring — including the on-device camera-AI layers (face liveness, phone detection, snapshot evidence) — and a risk-scored review dossier per attempt with recording playback and a snapshot photo strip; a grading workspace with AI-drafted feedback and "Draft grade with AI" per-criterion prefills — always editable, never auto-released; submission similarity screening with side-by-side excerpt comparison; anonymous mid-semester feedback windows with AI theme summaries; average peer ratings surfaced in grading; at-risk early warning with one-click intervention; one-click attendance; bookable office hours; section analytics; and discussion moderation in their own sections.

Administrators

Centralised control over users, roles and a fine-grained permission matrix (46 permissions, time-limited grants); departments, terms, courses, sections and rollover; course prerequisites and FIFO section waitlists with auto-promotion on drops; the document library and approval workflow; the exam-security gate governing which proctoring layers faculty may use, with a built-in step-by-step layer guide; exams and announcements; a discussion moderation queue; institution-wide analytics and top queries; a User Activity panel reporting visitors, referrers, devices and IP-derived locations with filters and configurable retention; AI settings and retrieval tuning with a live connection test; per-user AI usage monitoring with revocable access; a demo-mode AI usage cap; a hidden URL-driven maintenance switch; and a live system-health monitor.

Platform-wide

- **Document approval workflow:** every upload passes a review queue (approve, reject with reason, request changes, comment). Approval automatically extracts, chunks, embeds and indexes it into the AI's knowledge; rejecting or editing it removes it from answers instantly.
- **Layered exam proctoring (17 layers):** faculty pick, per test, from an admin-approved set — fullscreen, tab-switch detection, clipboard block, one-question-at-a-time, disable-back, question and option randomisation, identity watermark, integrity notice, browser fingerprint, behaviour logging, risk scoring, webcam and screen recording, plus on-device camera AI: a blink-verified face-liveness gate (questions stay hidden until a live face is confirmed; face loss blurs then locks the paper with escalating warnings), phone and second-face detection, and snapshot evidence (photo bursts at flagged moments plus random samples — moment-based evidence instead of hours of video). Every attempt produces an evidence trail and a 0–100 risk-scored dossier. Correct answers never reach the browser during a live attempt; grading is server-authoritative and instant.
- **Granular access control:** a visual role-and-permission editor, 46 fine-grained permissions, temporary roles that auto-expire, a full audit log, and a protected admin account that can never be locked out.
- **Semantic global search (🔍):** one search by meaning across documents, notes, materials, courses, assignments, discussions and past AI chats — scoped to what the user may access.
- **Notifications, digests and messaging:** system events reach exactly the right people; a Monday email digest and daily due-soon nudges (opt-out) keep deadlines visible; students and their own teachers message in real time with presence, typing indicators and unread counts, alongside a live system-health monitor.
- **Operational governance:** zero-latency page-view tracking behind an admin activity report, a demo-mode AI request cap protecting shared provider budgets, and an operator maintenance switch that can freeze or restore the platform from any URL — with the health check always reachable.

7. Impact and Feasibility

Technically, UniNexus is feasible and production-ready — not a concept. Auth and RBAC, the full document → chunk → embed → retrieve → cite → answer pipeline, personal-corpus RAG, the agentic tool layer, streaming, 17-layer proctoring with on-device camera AI, grading, analytics, messaging and admin governance are all implemented end-to-end across the three roles, backed by an automated test suite of 235 PHP feature tests

across 45 suites plus JavaScript component tests. Because there is no external vector database and no third-party LLM SDK — and because chat and embeddings both fall back automatically across providers down to a keyless offline one — the system runs on modest infrastructure and can be demonstrated on any laptop with no API keys at all.

The societal impact is significant. In a country like Bangladesh, where universities routinely struggle with resource fragmentation and inconsistent academic communication, a centralised AI-powered platform can meaningfully raise both the quality and the equity of education. Making institutional knowledge instantly reachable for every student — and cutting administrative friction for faculty — improves academic outcomes at institutions of any size.

Technologically, UniNexus demonstrates how RAG-based and agentic AI can be responsibly deployed in a high-stakes academic environment: every answer cited, every action permission-checked, every AI-drafted grade reviewed by a human before release, and every proctoring signal treated as evidence for human review rather than an automated verdict. It is a model other institutions can follow.

8. Scalability & Future Vision

UniNexus is built to scale. Its section-based, multi-tenant-ready architecture supports thousands of students, faculty, departments and documents, with an end-of-term rollover that winds classes down, locks in grades and promotes the next semester automatically. Heavy work — embedding, indexing, similarity screening, digests, evidence pruning — runs on queues, and the vector store is deliberately isolated behind a swap point, so a managed vector database can replace the in-MySQL store at scale without touching domain code.

Planned next: Telegram and WhatsApp alerts for assignments, quizzes and announcements, and a digital library with an AI assistant answering strictly from its own content. Voice interaction, lecture-audio transcription, completion certificates and expanded real-time presence are designed and intentionally on hold. Later phases include ML-driven predictive analytics building on the shipped rule-based at-risk signals, additional AI providers (Gemini and self-hosted models), document version history and longer conversation memory, and multi-institution support.

The long-term vision is to turn universities into AI-driven intelligent ecosystems, where academic knowledge is instantly reachable, administrative friction is minimised, and every AI action is grounded, cited and accountable.